

Python

born 1991

```
for n in range(3):  
    print(n)
```

Runs by

Interpreted, line by line

Good at

Reads almost like English. Huge libraries for data, AI, and automating chores.

The catch

Slower than compiled languages. Doesn't run natively on phones or in browsers.

Found in

Data science, AI, quick scripts, CMU CS Academy

JavaScript

born 1995

```
let score = 0;  
score++;  
document.title = score;
```

Runs by

Interpreted by the web browser

Good at

The one language every web browser runs. Makes pages interactive.

The catch

Quirky rules: "2" + 2 gives "22".

Found in

Nearly every website; also web servers (Node.js)

Java

born 1995

```
int x = 5;  
System.out.println(x);
```

Runs by

Compiled to bytecode, run by the Java Virtual Machine

Good at

"Write once, run anywhere." Strict rules help huge teams work together.

The catch

Wordy. Even tiny programs need a lot of setup.

Found in

Big company servers, older Android apps, Minecraft Java Edition, AP CS A

C

born 1972

```
int x = 5;  
printf("%d\n", x);
```

Runs by

Compiled to machine code

Good at

Fast and tiny. Works very close to the hardware.

The catch

You manage memory yourself. Mistakes cause crashes and security holes.

Found in

Operating systems, microwaves, thermostats, car computers

C++

born 1985

```
int x = 5;  
std::cout << x;
```

Runs by

Compiled to machine code

Good at

C's speed plus tools for organizing very large programs.

The catch

Enormous and complicated. Easy to make subtle mistakes.

Found in

Game engines (Unreal), web browsers, video editing software

C#

born 2000

```
int x = 5;  
Console.WriteLine(x);
```

Runs by

Compiled to bytecode, run by Microsoft's .NET

Good at

Clean, modern, well supported. The scripting language of Unity.

The catch

Strongest inside the Microsoft and Unity worlds.

Found in

Unity games, Windows software, business apps

Swift

born 2014

```
let name = "Ada"  
print("Hi, \(name)")
```

Runs by

Compiled to machine code

Good at

Apple's modern language. Safe, fast, and built into Apple's tools.

The catch

Mostly used only on Apple devices.

Found in

iPhone, iPad, Mac, and Apple Watch apps

Kotlin

born 2011

```
val name = "Ada"  
println("Hi, $name")
```

Runs by

Compiled to bytecode, like Java

Good at

Google's preferred Android language. Shorter than Java and works alongside Java code.

The catch

Mostly used for Android and Java-world projects.

Found in

Android apps

SQL

born 1974

```
SELECT name FROM pets  
WHERE age > 3;
```

Runs by

Inside a database. You describe what you want, not how to get it.

Good at

Asking questions of enormous tables of data, fast.

The catch

Not general-purpose. You can't build a whole app with it.

Found in

Behind almost every app that stores data

R

born 1993

```
scores ← c(88, 92, 75)
mean(scores)
```

Runs by

Interpreted

Good at

Statistics and graphs. Built by statisticians, for statisticians.

The catch

Unusual syntax. Not meant for apps or games.

Found in

Research labs, biology, medicine, public health

Rust

born 2015

```
let x: i32 = 5;
println!("{}", x);
```

Runs by

Compiled to machine code

Good at

As fast as C, but the compiler blocks whole categories of memory bugs.

The catch

Hard to learn. The compiler is very strict.

Found in

Parts of web browsers and operating systems, security-critical tools

Go

born 2009

```
x := 5
fmt.Println(x)
```

Runs by

Compiled into a single file

Good at

Simple, fast, and great at doing many tasks at once. Easy to deploy.

The catch

Leaves out features on purpose, so some code gets repetitive.

Found in

Cloud services and web servers

Scratch

born 2007

when green flag clicked

say Hi! for 2 seconds

Runs by

Snapping blocks together in a web browser

Good at

No syntax errors possible. Instant animations and games.

The catch

Hard to build large or professional software.

Found in

Classrooms and first-time coders, ages 8 and up

Assembly

born 1940s

```
mov eax, 5
add eax, 2
```

Runs by

Translated almost directly into the processor's own instructions

Good at

Total control of the chip. Nothing runs faster.

The catch

Each chip family has its own. Slow to write, hard to read.

Found in

Startup code for chips, tiny devices, speed-critical pieces

PHP

born 1995

```
<?php
echo "Hi, visitor!";
?>
```

Runs by

Interpreted on a web server, which sends back finished pages

Good at

Building web pages on the server. Cheap, easy hosting everywhere.

The catch

A messy history of inconsistent design.

Found in

WordPress (a huge share of all websites), Wikipedia

Lua

born 1993

```
local coins = 0
coins = coins + 1
```

Runs by

Interpreted, usually inside another program

Good at

Tiny and easy to plug into games and apps as a scripting language.

The catch

Very small toolkit. Rarely used on its own.

Found in

Roblox (as Luau), game mods and add-ons

COBOL

born 1959

```
ADD 1 TO COUNTER.
DISPLAY COUNTER.
```

Runs by

Compiled, usually on big mainframe computers

Good at

Business record-keeping. Reads like English. Rock-solid for decades.

The catch

Few new programmers learn it.

Found in

Banks, insurance companies, government systems

HTML/CSS

born 1993/96

```
<h1>Hi!</h1>
h1 { color: teal; }
```

Runs by

Read and drawn by the web browser

Good at

HTML gives a page its structure. CSS gives it its look.

The catch

No decisions and no loops on its own.

Wild card: is this a programming language? Argue it.

The job

A

Your school wants an iPhone app that shows today's lunch menu.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

B

A local gym wants an Android app where members log their workouts.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

C

A teacher wants a quiz that runs in any web browser, with buttons and a score counter. Students can't install anything.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

D

A 3rd-grade class wants to make animated stories about their favorite animals.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

E

A friend needs a portfolio page for their art. It just has to look good: no buttons, no logic.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

F

An online store's data lives in a database. They want a list of every customer who ordered more than 3 times last month.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

G

A smart thermostat has a tiny chip with very little memory. The code has to fit and run for years without crashing.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

H

A big studio is building a 3D game engine that has to hit 120 frames per second.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

I

A two-person team is making an indie game in the Unity game engine.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

J

You want to add custom scripts to your Roblox obby, like a door that opens when you collect 10 coins.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

K

A bank's account system has run since 1985. There's a bug to fix by Friday, and rewriting the system is not an option.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

L

A web browser team needs new code that is fast and can't have memory bugs that hackers could exploit.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

M

A startup's cloud service has to handle thousands of requests at the same moment and be easy to deploy.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

N

A company with 500 developers needs a backend system that runs on any server and lasts 20 years.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

O

An engineer has to write the very first instructions a brand-new chip runs when it powers on.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

P

A small bakery wants a contact form added to its existing WordPress website.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

Q

You need to rename 2,000 field-trip photos by date. You need it done this afternoon, and you'll only do it once.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

R

A biology lab needs statistical tests and publication-quality graphs for a research paper.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

S

A pacemaker's software keeps a patient's heart beating. If it crashes, the patient could die.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

T

A science fair robot needs to drive in a circle and blink its lights during a 5-minute demo.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

U

A car's automatic emergency brakes have to react within milliseconds, every single time.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

V

A Discord bot for your friend group that posts a meme every morning.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

W

A stock-trading program where being one millisecond slower than competitors loses real money.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

X

Managers get a sales report emailed every morning. It runs overnight, so taking an hour is fine, but it has to arrive.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

Y

A hospital system tracks which medications each patient gets and when.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

The job

Z

A Mars rover lands itself 140 million miles away. Nobody can walk over and press restart.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because:

Your job

?

Write your own job, then trade it with another team.

☐ Mission critical ☐ Must be reliable
☐ OK to fail or be slow

Our pick:

Because: